

Visilock – Face recognition Based Door Lock System

Group No - 53

Rahul Kumar Das – 24075102

Trishan Roy -24075107

Umer Sagri - 24074036

1. ABSTRACT

The rapid growth of **smart home appliances** has led to a huge demand for **secure and automated door lock systems**. Traditional **key-based** and **password based door lock systems** are vulnerable to *theft, duplication and forgetting*. This is where **our project** steps in.

Our project provides a **low cost, IoT-based face recognition door lock system** integrated with **Telegram**. This helps in *remote monitoring and control*. Our door lock system *captures the image of a person* through the **camera**, performs **face detection and recognition** based on *saved dataset* using **ESP32-CAM's inbuilt processing capability**.

If the face matches , the **relay is triggered** to *unlock the door for 10 seconds* and the user is also *notified* with a **known face detected message**. In case of *failure*, **unauthorized access** is also reported immediately to the owner with an **unknown face detected message** along with a *photo of the intruder*.

The system operates over **Wi-Fi** with *real-time communication* and *remote operation*. The results of the project demonstrate **high accuracy, fast response and enhanced security** compared to traditional door locks. Our project works well under *moderate lighting conditions*. Also when a *photo of the intruder* is taken, the **flash mode of the ESP32-CAM** is efficiently used.

Our project provides a cost-effective, reliable and **user friendly solution** for smart home appliances.

2. Background Of The Project

Introduction :

- With the expansion of **IoT based systems and technology**, *automation* is employed in a considerable number of modern-day home appliances. There is a need to completely reform traditional measures in **security check and access control** spaces. Conventional models like *conventional door locks* are simple but have several *difficulties* in maintaining **physical entrance security**. One may lose a door key, and someone else may *generate a carbon copy* of it. Passwords can be *forgotten*.
- Thus, **Smart door locks** have been devised for this purpose. They offer a *simple interface* and *strengthen security* at a **low price** in an *easy way*. They can also be *enhanced* with other accessories to tighten security around the door.

Problem statement :

- The elementary, ordinary systems like verified *keys*, *verified RFID Tags*, or the *regular pressure pad*; homeowner monitoring securities are the prevalent ones. *Stealing* these devices is still open; for instance, the keys can be **missed or duplicated**, and the fob can be *misplaced*, and sometimes the user can *lose the password*. There is a lack of an imperative, which is a lock that would *verify a person's identity* and can be **operated from a distant** .

Existing Solution :

- **Smart locks** are surrounding us nowadays and operate based on 3 systematic principles: *fingerprinting, RFID- readers identification or Numeral keypad*. The key or tag can be scanned with **IoT-based devices** with **ESP32-CAM** circuitry and be a safety wallet tool, but **conditional triggers require servers for processing** and are not automated.
- The **core server** can be built on for *real-time notifications* of data getting updated to generate notifications to the homeowner user... For that, the system should be **face recognition robust**, continue to update feed, and simple lock control through the smartphone that makes it best for each **home or workplaces or restricted area**.

3. PROJECT PURPOSE

The purpose of the project is to develop :

- ❖ A *face-recognition* door lock system designed and implemented with an **ESP32 camera module**.
- ❖ A **door-critical system** that can detect and identify faces using the **ESP32 camera** *at all times without needing additional hardware*.
- ❖ *Increased security* of a door lock to develop a device where **the outsider cannot unlock unless the owner grants access**.
- ❖ A *safe registration mechanism* that **registers users and verifies them** when attending on the *hosted* server.
- ❖ Driving of a **lock mechanism** controlled and regulated using a **relay module**.
- ❖ The development of a *tiny* and **low-cost system** that *functions in place of a big or intricate mechanical and electrode lock*.
- ❖ **Lowered the price** of a smart lock to be *affordable* to a majority of people who buy a home.
- ❖ An *easier to install system* with a **user-friendly interface** makes *adding new users or removing ones easy and secure*.
- ❖ A **remotely controlled system** that allows a user to *access the lock system via an app from a distance*.
- ❖ **Telegram** used to *send lock, unlock commands, alerts, and remote accessing* of the system.
- ❖ A system that operates on **moderate light** and does not disturb *accuracy*.
- ❖ **Low power consumption** for *maximum battery efficiency*.
- ❖ A **homeowners' system** which is *affordable* to install in an **office or houses**.

4. NOVELTY IN THE PROJECT

- The proposed system has several **novel and pioneering features** that separate it from the conventional door lock systems and their expensive replacements.
- Firstly, the proposed system uses the **ESP32-CAM module**. It is a *cheap and small IoT microcontroller* that supports both **WiFi** and **Bluetooth**, which are compatible with *image processing, video streaming, and security systems*. Additionally, it has a **built-in microSD card slot** for storage.
- **Facial identification and recognition** are performed on the **ESP32-CAM**, which *greatly reduces our hardware costs and power demands*.
- Our project is also built with the help of the **Telegram Bot API**. It makes the telegram app possible for all of the users to get *live notifications, images taken at a worksite, and remote control*, which help the owner to *close/unlock* the lock on the door at a time. Also, the owner gets an alert whenever an **unauthorized person** enters the cabin with *photographs of the person*. **IoT communication** and **on-device intelligence** is a novel function in the proposed system.
- Unlike in expensive door lock systems, our project is quite **inexpensive and user friendly**. Existing systems rely on *external servers or expensive biometric modules*. Whereas our project provides a **standalone, low cost and scalable solution**.
- In traditional systems which use *pin based door locks*, there is a high chance of forgetting the *pin* or the system getting *hacked*. But in our project this problem is solved by using **face recognition**.

5. SUMMARY OF THE PROJECT

The **Face Recognition Door Lock System with Telegram Integration** is an essential **smart home and office IoT-based project** to help provide the *security enhancement* needed in one's living or work environment.

It involves the **ESP32-CAM module**, the **relay module** which triggers the door lock, and the **indicator LEDs** that are put off when the access is denied due to *unauthorized intrusion*. The **ESP32-CAM** takes a *snapshot* when it detects a person, then processes the image taken onboard where the actual **face recognition algorithm** is embedded.

If the identified face is *registered*, the *relay opens* the door lock and holds it for **10 seconds** as the indicator LEDs blink. If it is an *unrecognized person*, the relay activates the camera to take a *snapshot of the intruder* and deny access.

The **secure system architecture** includes the **Wi-Fi interface** suitable for both *monitoring and controlling* from any geographical location worldwide.

Telegram is also integrated to offer a *secure user interface*, *receive photos* and *bot updates*, and control *the door lock remotely*.

The system is **programmed using the Arduino IDE software** and tested under various conditions. The **Telegram Bot** supports the following 5 commands :

- **/start** – *Shows the command description for this unit*
- **/photo** – *Takes and sends the actual camera image*
- **/unlock** – *Unlocks the door through the relay and holds it for 10 seconds*
- **/lock** – *Locks the door immediately*
- **/status** – *Gives the actual system status*

Final Result : The result obtained from the project indicates that the designed system implements **fast recognition, direct face detection** and **remote access** which is secure with *24/7 monitoring*.

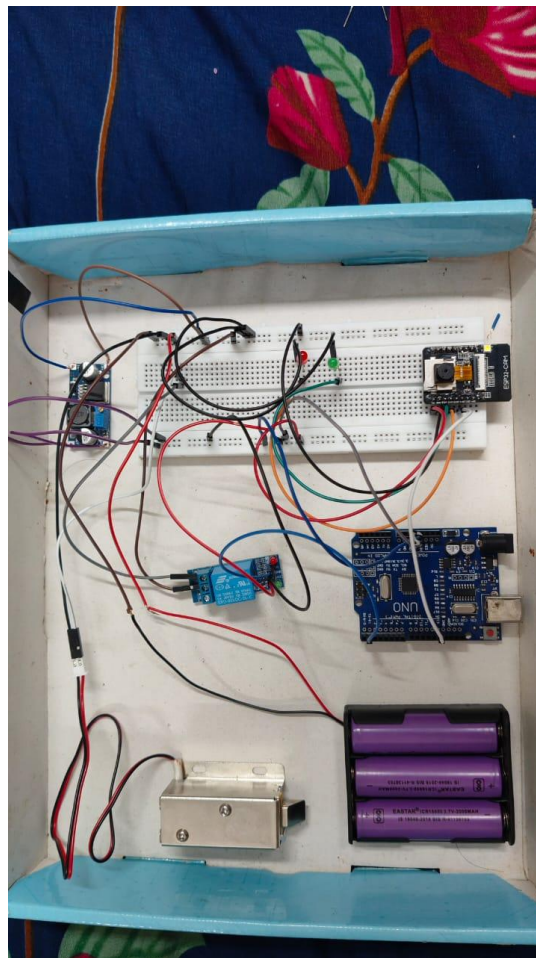
Conclusion : The smart door lock indicates that **ESP32-CAM microcontroller** can provide *real-time face recognition* and **IoT attainments at low cost**. The final system incorporates the **IoT security, automation and reachability at the best low-cost solution package**.

6. LIST OF DRAWINGS WITH BRIEF DESCRIPTION

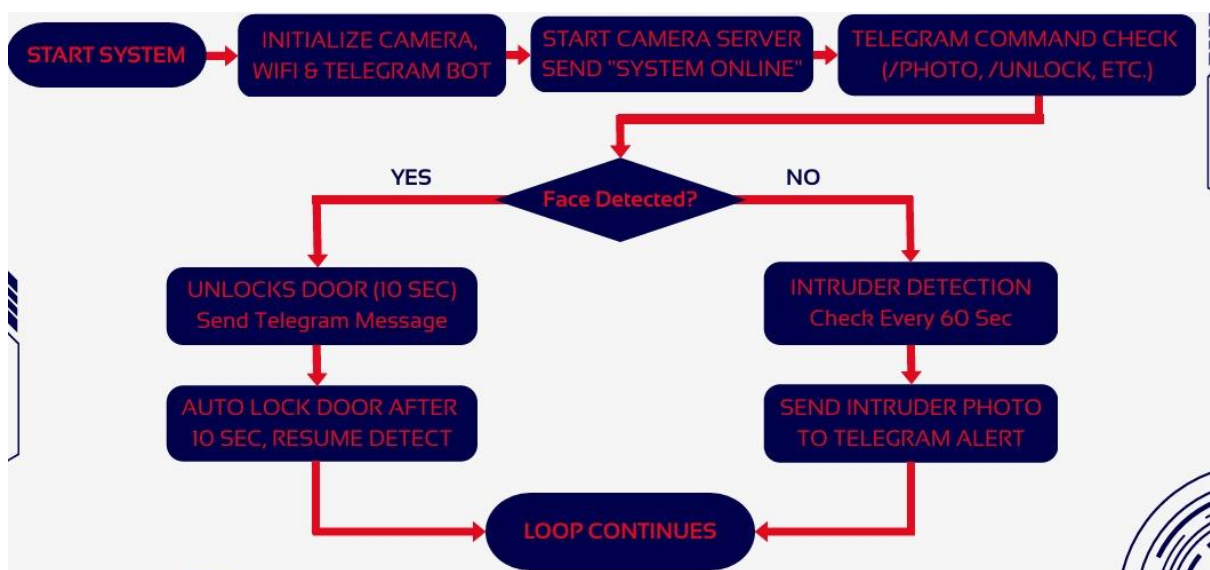
- **ESP32-CAM module** : It is a module that includes a *camera*, a *microSD storage card slot*, and the *ESP32 microcontroller*. This *low-cost development module* supports both *Wi-Fi and Bluetooth connectivity* and is an *excellent component of IoT solutions*. *Image processing, video streaming, or security systems* are just a few examples of use-cases.
- **Relay Module** : *Relay Module* works as the *switch to turn on and off electric lock mechanisms*, which are the *solenoid locks*. When the *ESP32-CAM module* detects the *face*, it knows and *sends an electric signal to the relay module*.
- **Solenoid lock** : A *Solenoid lock* is an *electromechanical lock* that *responds to electric current*. It contains a *metal pin* that moves in a *horizontal or vertical manner* to control the *lock-open and lock-close*.
- **Buck Converter** : *Buck converter* is a *DC-DC converter* that *receives 12V input* and after *conversion* gives the *output voltage and current of the exact value required by the ESP32-CAM and all the components*.

- **Arduino UNO** : It is a *microcontroller board* that is *small-sized* and mainly used as an *interface to step-up the 3.3V being obtained from the ESP32-CAM to about 5V* before it is *distributed to the relay module*.
- **Power supply** : It consists of 3 *rechargeable cells each with 3.7V* cells added up to provide a *total of 12V* to all the circuits.
- **LED Lights (Red and Green)**: Acts to *visualize the status of the lock being open (green) and closed (red)*.
- **Telegram Bot API** : *Telegram bot that sends notifications, images, and allows controlling the solenoid lock remotely using several commands*.

CIRCUIT DIAGRAM:



BLOCK DIAGRAM(FLOWCHART):



7. DETAILED DESCRIPTION OF THE PROJECT

The system's architecture : The system's architecture encompasses multiple hardware components realized with the **ESP32-CAM**, a **relay**, **LEDs** besides the software developed with the *Arduino IDE*, and the *Telegram API*. The **ESP32-CAM** serves as the system's brain which encapsulates *facial captures, facial detection and recognition, stemming, and communication over Wi-Fi*. Contrastingly, the **Telegram Bot** manifests as an *IoT-based graphical user interface platform* leveraging the computer-based through image detection.

Working Principle : The **Face Recognition-Based Door Lock System** operates around the *Computer vision* and the *Internet of Things* for the door access control and remote monitor. This system's core is the **ESP32-CAM module**, which oversees the *image detection and recognition part*. Upon on, the **ESP32 microcontroller** kicks in with the activation of its *camera module* and the *Telegram Bot* with the Wi-Fi connected. It starts to capture video frames and span the analysis against the person by employing the *face detection and recognition algorithms*. The *face detection algorithm* seeks for human faces in its viewing angle on the camera based on the face patterns like eyes and mouth etc. The *face recognition algorithm* against some batches of images stored in a database and classifies them automatically. **If matches, the person is approved and meanwhile, the door will be opened.** Additionally, it integrates the **Telegram Bot**, which mandates a user receiving the *lock/unlocking notifications* and some *commands functioning as the remote control* of the door lock. This context ensures the *intelligent and secure system* of door access.

Methodology : The system development methodology comprises the *development of hardware and software, integration and testing* of the system. Each stage of the system is necessary to ensure system functioning and a reliable real-time operation system. **After powering ON the ESP32-CAM high resolution camera module, it automatically connects to the pre-configured Wi-Fi network** and is ready to turn on. As a system integrates, it starts the *camera module* and the *Telegram Bot API*. A **confirmation message with ‘ESP32 CAM Door Lock System Online’** is sent to the system link telegram chat.

Face Detection and Recognition : Once the user opens the *Face Recognition Interface* through the **ESP32-CAM’s static IP address** on his laptop or smartphone browser. The Interface commands such as: *start stream, face detection, sample, face recognition*, and then have optional like *V-flip, change camera window size* etc. If they start streaming, while the face is displayed on the interface, the *sample option* then stores that face the face owner logged. Their faces with the *Eigenfaces* are recognized with the existing samples in the uploaded data. **If the recognized face safely exists matched data, he can access it, and when the faces aren’t matching, the Doorlock system remains locked** according to the sample. Again starting when the face has screenshots, the **relay module will remain triggered**, and the **door will remain unlocked**. When 10s the door will be autolocked whenever the face is not identified then remains locked.

Telegram Integration : The **ESP32-CAM** further sends a “*System Online*” message to the *Telegram Bot API* on the first Wi-Fi connection and, at the same time, passes a set of commands. The **Telegram Bot** is given control and feedback commands are as follows: */start- shows list of commands available to the bot, /photo-Captures, and sends the current camera image, /unlock- unlocks the door remotely for 10 seconds, /lock- locks the door. Immediately, /status- reports the system status* when the system executes is as designed. During the system. **A photo is taken and sent every 240**

seconds until an authorized user is recognized as such. In the latter case, a *confirmation message* is sent to the Telegram chat.

System Feedback : When an *authorized face* is recognized, the **green LED** is turned on for 10 seconds, and door blocks. In case an *unknown person* is detected at the door, the **red LED** remains on. Moreover, if the door blocks, the red LED also remains on.

Hardware Implementation :

- **ESP32-CAM:** Acts as the *brain* and handles *face detection and recognition*.
- **Relay Module:** Connected to *GPIO pin 15* and triggers the *electric lock*.
- **LEDs:** Indicates access/door status (*Green = Unlocked, Red = Locked*).
- **Solenoid Lock:** Acts as the *door lock*.
- **Power Supply:** *12V source* using 3 rechargeable batteries.
- **Power converters:** *5V regulated source* for ESP32-CAM using *Buck Converter* and *3.3V from ESP32-CAM step upped to 5V using Arduino UNO board*.

Software Implementation :

- Developed and coded using *Arduino IDE*.
- Several libraries have been used which include *esp_camera.h, WiFi.h, UniversalTelegramBot.h, ArduinoJson.h* etc.
- The **ESP32-CAM** is programmed to use *Telegram's bot token and Chat ID* for communication.
- The *face recognition model* uses the *internal library functions* of the ESP32-CAM module.

8. RESULT

Current Outputs :

- Our **working prototype** has successfully implemented **face recognition** and **Telegram based remote control**.
- The relay is **verified** – the door unlocks during tests correctly.
- The **status LEDs(Red/ Green)** are working properly.
- **Auto-relock after 10 seconds** is working.
- The **Intruder Detection and alert system** is working and sends a message whenever an intruder is detected.
- **Manual commands** using **Telegram Bot** - *"/photo, /status, /lock, /unlock"* are working.
- **Wifi and Telegram Connections** are consistent for long durations. Under *well-lit conditions and low luminance (100Lx)* it is good under spectacle wearing.

Known Observations :

- Telegram photo transmission may fail/ delay in weak network conditions.
- The Face Recognition is not accurate in less luminance, but it can be enhanced by the "Flash LED".

